

Hemant Ghuge

📍 Pune, India | 📞 +91 771 808 5056 | 🌐 [AI Portfolio](#) | [in hemantghuge](#) | [G hemantghuge@gmail.com](#)

PROFESSIONAL SUMMARY

AI Engineer with 4.5+ years of experience architecting production-grade ML systems, focused on computer vision, edge deployment, and emerging GenAI workflows. Proven track record of delivering scalable DL pipelines powering real-time safety and analytics across 9,000+ ATMs and 1,200+ retail stores. Proficient in RAG pipelines with hybrid search, lightweight VLM experimentation (ByteVLM), and prototyping with OpenAI APIs, LangChain, and FastAPI. Skilled in orchestrating 1,500+ edge devices via Tarsyer Bridge, a remote management platform that enables secure access and control of AI devices deployed across PAN India, USA, Indonesia and South Africa. Passionate about translating business needs into AI-first solutions with a focus on performance, reliability, and user value.

WORK EXPERIENCE

- Tarsyer Insights LLP**  **Pune, India**
◦ *Sr. AI Engineer | Acting Product Lead* Aug 2020 – June 2025
Joined as Intern → DL Engineer → Sr. AI Engineer
Tarsyer began as a 3-member startup focused on AI-powered insights for manufacturing and security. Over time, our work evolved to focus on the retail and logistics sectors. I joined in the early phase and played a central role in scaling both the team and product offerings - helping grow Tarsyer into a multi-crore AI product company with nationwide and global deployments across retail, logistics, and public safety domains.
- Deployed GenAI prototype **Pickup Assist** using **Vertex AI**, enabling visual/text reasoning workflows optimized for delivery agents.
 - Built and deployed **ByteVLM**, a compact vision-language model running on **Raspberry Pi**, enabling efficient edge inference for image-text tasks.
 - Designed a **hybrid RAG system** using **Weaviate** with combined vector and keyword search, improving visual query relevance in image discovery tools.
 - Built a **centralized dashboard system** from scratch to support 1,200+ users across retail stores, integrating **device status indicators**, and **configured camera images** for enhanced operational visibility.
 - Developed a **custom object detection training pipeline** and **lightweight tracker** optimized for **real-time, on-device performance** - enabling reliable detection and tracking across edge-deployed surveillance and monitoring applications.
 - Delivered and scaled **production-grade DL/CV systems** for 9,000+ ATMs and **1,200+ retail stores**, enabling **helmet detection**, **loitering alerts**, and **footfall analysis**.
 - Designed and deployed a **Driver Monitoring System** with **6+ AI features** (e.g., eye closure, phone usage, drowsiness), optimized to run under **\$110 total cost** per unit, balancing accuracy and affordability for transport clients.
 - Spearheaded development of Tarsyer Bridge, a remote orchestration platform to access and manage 1,500+ edge devices deployed across India and **international sites**: enabled **monthly cost savings exceeding \$1,200**.
 - Collaborated with **QA and client teams** to validate deployments and improve post-deployment workflows across diverse environments.

PROJECT IMPACT HIGHLIGHTS

- **Generative AI (Pickup Assist)**: Built a Vertex AI-powered assistant to aid last-mile delivery workflows.
- **ByteVLM + Raspberry Pi**: Deployed compact **vision-language model** on RPi for local visual reasoning workflows.
- **Retail Analytics (Trent, Bata)**: AI-based **footfall tracking** across **1,200+ stores** for business optimization.
- **ATM Safety Monitoring (SBI, HDFC)**: Deployed helmet and loitering detection models across 9,000+ ATMs nationwide.
- **AI Cement Bag Counter**: India's first **AI warehouse bag counter system** using edge ML; reduced manual error.

EDUCATION

- Government College of Engineering and Research**  **Pune, India**
◦ *Bachelor of Engineering in Electronics and Telecommunication, CGPA - 7.97 (First Class with Distinction)* Aug 2016 – Nov 2020
- **Leadership & Mentorship**: Led 18-member team for tech fest; conducted Raspberry Pi & MATLAB workshops
 - **Research**: Published 3 international research papers at college

SKILLS

- **AI & ML**: LLMs · VLMs · GenAI · RAG · Computer Vision · TensorFlow · OpenCV · Deep Learning · Edge AI
- **GenAI**: Vertex AI · OpenAI API · LangChain · Prompt Engineering · Hybrid Search · Weaviate
- **Engineering**: Python · REST APIs · Cloud · CI/CD · GitHub Actions · Docker · Streamlit · FastAPI · Git · GitHub
- **MLOps**: Raspberry Pi · Real-time Inference · Model Evaluation · Model Deployment · PRD Writing · Edge Optimization
- **Cross-functional**: Product Ownership · Customer Discovery · International Client Communication · Agile Workflow